# Vehicle Architecture-Driven Design and Comprehensive Loss Analysis of an 800V Three-Level Flying Capacitor Traction Inverter

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# Project Overview

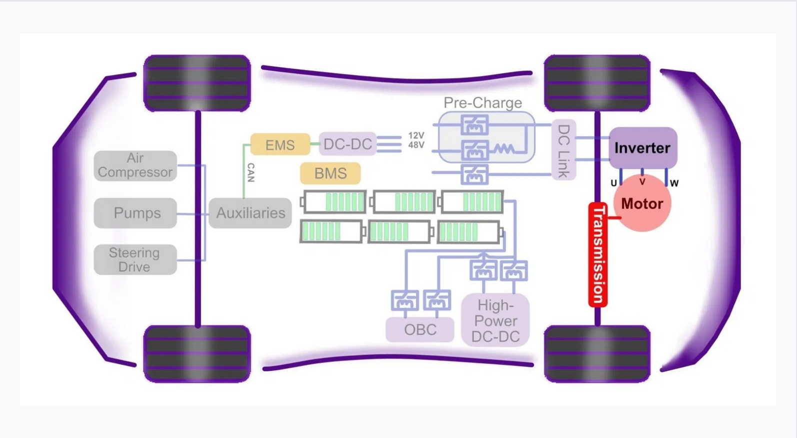
- **The Core Problem:** Traction inverter design requires balancing efficiency, torque quality, and thermal stress across highly variable driving conditions.
- **Variable Frequency:** The required electrical fundamental frequency ($f_e$) is not fixed; it is directly proportional to the vehicle's mechanical speed.
- **Load Profile:** Urban driving demands low frequency but high phase currents for acceleration. Highway driving demands high frequency but lower sustained currents.

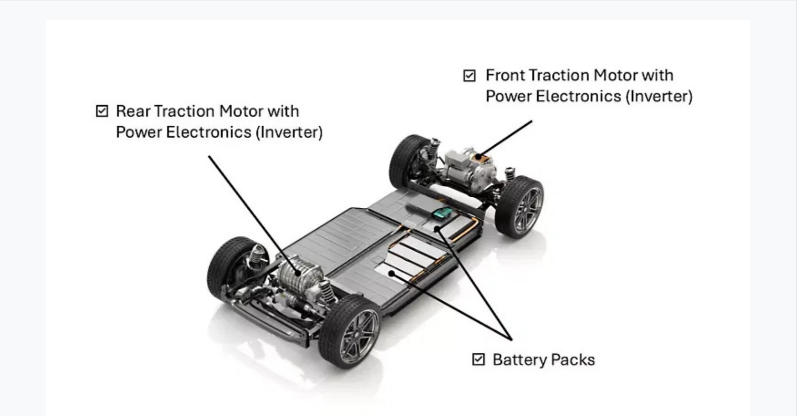
## Project Objective

To model a traction inverter across a full spectrum of operating conditions (urban to highway), evaluate comprehensive conduction and switching losses, and optimize the design for maximum drivetrain efficiency and thermal reliability.

# Design Choice: 800V Architecture

- **The Baseline Decision:** To address the performance demands of modern EVs, an 800V DC bus architecture was explicitly chosen as the baseline for this design project.
- **Current Reduction:** Adopting an 800V architecture halves the required phase current for a given power level compared to traditional 400V systems.
- **Efficiency Gains:** This drastic reduction in current significantly lowers $I^2R$ ohmic losses in the inverter semiconductors, motor windings, and high-voltage cabling.



# The dv/dt Problem and Multilevel Solution

- **The 1200V Requirement:** Traditional two-level voltage source inverters (2L-VSI) require 1200V-class semiconductors to operate on an 800V DC bus.
- **The SiC Trade-off:** While Silicon Carbide (SiC) devices easily meet this voltage requirement and offer low losses, their extremely fast switching transitions generate severe **dv/dt stress**.
- **Motor Degradation:** This high rate of voltage change over time leads to premature motor winding insulation failure and bearing degradation.

## The Multilevel Solution

- To mitigate dv/dt stress, a **multilevel inverter topology** is required.
- It synthesizes the output voltage from multiple discrete levels, inherently reducing the voltage step size (e.g., 400V steps instead of 800V steps).
- This cuts the dv/dt stress in half and significantly improves the harmonic profile of the output waveform.

# Topology Selection: FC vs NPC

## Thermal Symmetry & Component Count

- **Asymmetric Stress in NPC:** NPC suffers from unequal loss distribution between inner and outer switches, complicating cooling system design.
- **Symmetric Stress in FC:** FC distributes voltage and current stresses symmetrically across all four switches. Every switch blocks exactly half the DC bus voltage (Vdc/2).
- **Reduced Conduction Loss:** FC eliminates the need for clamping diodes required in the NPC, removing components from the conduction path.

## Dynamic Load Balancing

- **Transient Instability in NPC:** The NPC neutral point is notoriously difficult to keep balanced under highly dynamic, rapidly fluctuating EV loads.
- **Natural Balancing in FC:** FC utilizes redundant switching states to naturally balance the flying capacitor voltage.
- **Power Factor Independence:** By actively selecting between redundant states based on the phase current direction, the FC remains balanced at Vdc/2 regardless of the load power factor.

# FC Switching States and PD-PWM

- **3-Level Output:** Synthesizes $+V_{dc}/2$, 0, and $-V_{dc}/2$ using four switches per phase, reducing voltage steps and dv/dt stress.
- **Redundant Zero States:** The '0' voltage level can be achieved by two different switch combinations, providing a degree of freedom in control.
- **Natural Balancing:** Phase-Disposition PWM (PD-PWM) naturally balances the flying capacitor voltage without complex closed-loop control.
- **Mechanism:** By alternating between redundant zero states based on the phase current direction, the capacitor charges or discharges to maintain exactly $V_{dc}/2$.

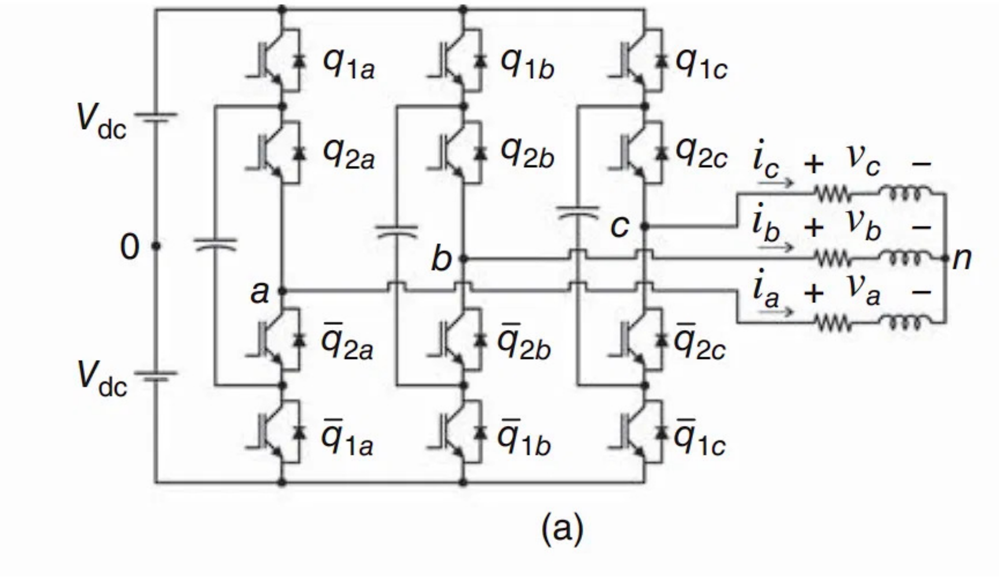



Fig. 1. Three-phase schematic diagram of the 3-level flying capacitor traction inverter.
Source: Advanced Power Electronics Converters (E. Cipriano).

# Component Selection

## Active Switch: SiC Module

- **Wolfspeed CAB450M12XM3:** 1200V, 450A SiC half-bridge module.
- **Voltage Margin (1200V):** Each switch blocks $V_{dc}/2$ = 400V; 33% voltage utilization per IPC-9592B derating.
- **Current Margin (450A):** ~3.3x margin vs. 136A peak phase current.
- **Thermal Stability:** Positive temp. coeff. of $R_{ds(on)}$ (die 2.6 + pkg 0.7 = 3.3 mΩ at 25°C, ~2× at 175°C) aids current sharing.

## Flying Capacitor: Film Tech

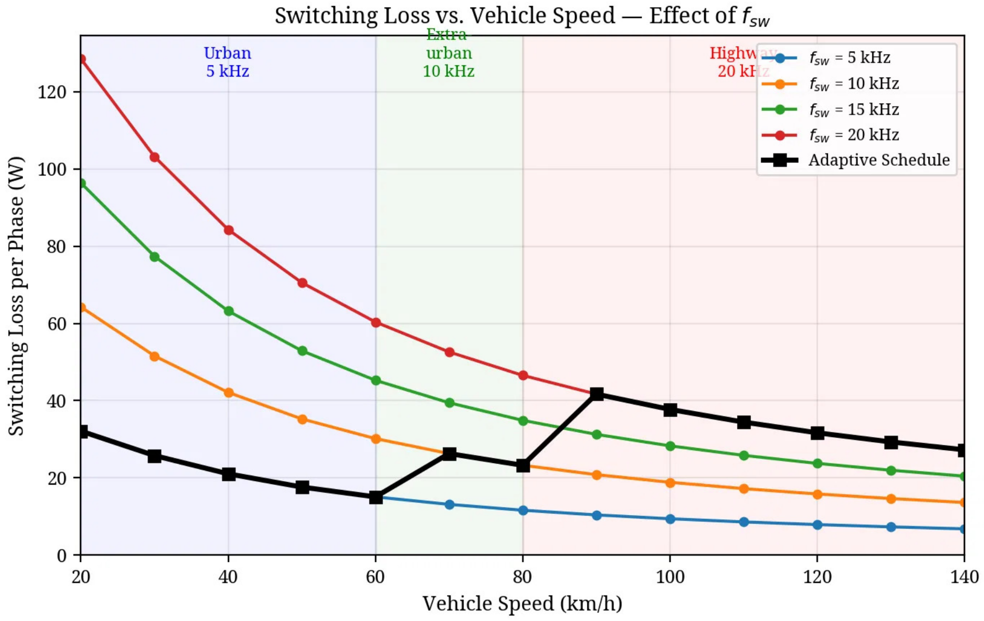
- **Capacitance (680 µF):** Keeps ripple within 5% (20V on 400V) at 5 kHz min switching.
- **Film vs. Electrolytic:** Automotive-grade metallized polypropylene chosen.
- **Key Advantages:** Very low ESR (<5 mΩ), long life, handles high-frequency ripple.

# Vehicle Architecture Mapping

- **Drivetrain Parameters:** Single-speed reduction gear (G = 8.19), tyre radius (r = 0.315 m), and 8-pole PMSM (p = 4).
- **Fundamental Frequency ($f_e$):** Directly proportional to vehicle speed.

$$f_e = (v / r) \times (G / 2\pi) \times p$$

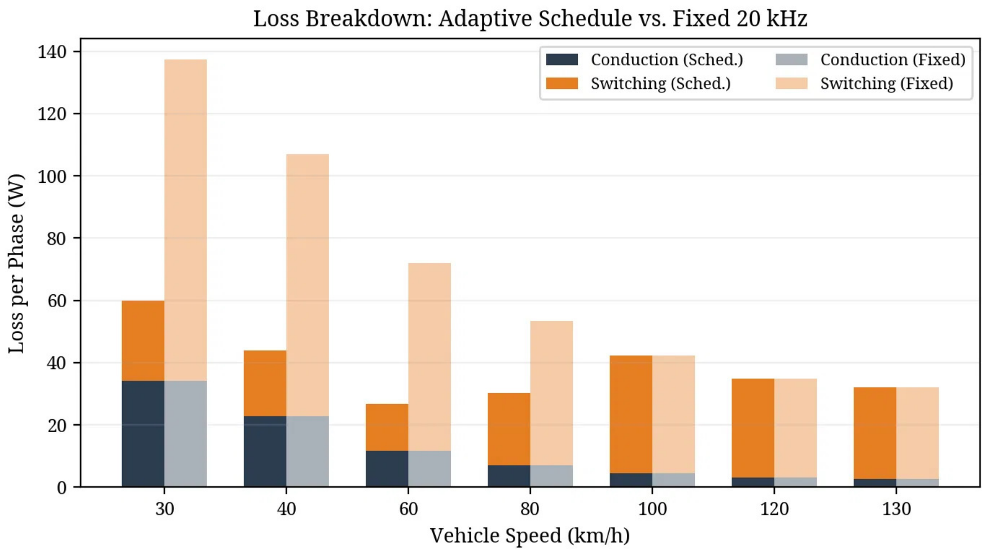
- **Urban Driving (40 km/h):** Low frequency (184 Hz) but high phase current (136 A) for acceleration. Represents the highest current stress.
- **Highway Cruising (120 km/h):** High frequency (552 Hz) but significantly lower sustained current (51 A) due to increased load impedance.



# Switching Loss Analysis

$$P_{sw} = N_c \times (f_{sw} / \pi) \times E_{sw\_ref} \times (I_{pk} / I_{ref}) \times (V_{dc\_sw} / V_{ref})$$

- **Graovac-Purschel Method:** Analytical method to capture drive-frequency effects.
- **Applicability to SiC:** Topology-dependent math; works for SiC by using device curves.
- **Datasheet Calibration:** Uses CAB450M12XM3 parameters (Eon = 25.4 mJ, Eoff = 7.51 mJ @ 450A/600V).
- **Fixed 20 kHz Issue:** At 20 kHz switching losses dominate at low speeds due to high phase currents (~136A).

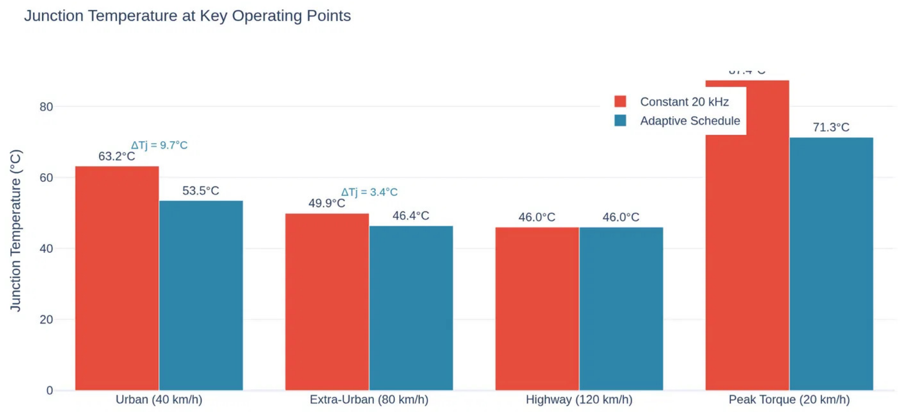


# Simulated Adaptive Switching Frequency

An adaptive switching frequency schedule was developed and fully implemented within the time-domain simulation. The core methodology relies on maintaining a sufficient frequency modulation ratio ($m_f = f_{sw}/f_e$) to ensure waveform quality while minimizing losses.

## Urban Driving (Low $f_e$)

Because $f_e$ is low (184 Hz), the controller dynamically reduces $f_{sw}$ to 5 kHz while maintaining a high modulation ratio ($m_f \approx 27$). This reduces switching losses by 75% at the operating point where phase current is highest (136A).



# Thermal Impact of Adaptive Schedule

- **Thermal Modeling:** Junction temperature ($T_j$) was evaluated using a 4-layer Foster thermal network derived directly from the CAB450M12XM3 datasheet.
- **Steady-State Reduction:** The adaptive schedule reduces $T_j$ by 16.2°C during peak torque conditions (20 km/h) and by 9.7°C during urban driving compared to a fixed 20 kHz baseline.
- **Transient Thermal Cycling:** Over a dynamic drive cycle, the adaptive schedule reduces the thermal cycling amplitude ($\Delta T_j$) by 53% (from 16.4°C to 7.7°C).
- **Reliability Gain:** Per the Coffin-Manson fatigue model, this 53% reduction in $\Delta T_j$ extends the power module's fatigue lifetime, as cycles-to-failure scales with $(\Delta T_j)^{-n}$ where n ≈ 4–5.

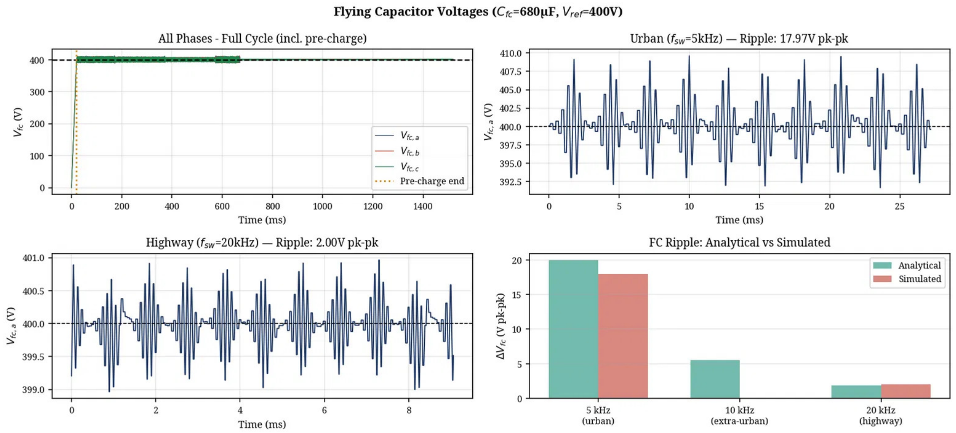



Fig. 4. Time-domain simulation validating FC voltage balancing, a prerequisite for stable thermal operation.

# Simulation Results

## FC Voltage Balancing

- **Pre-charge:** A 20 ms linear ramp prevents inrush currents during startup.
- **Natural Balancing:** PD-PWM maintains the flying capacitor voltage at 400V with a mean error < 0.04V.
- **Ripple Budget:** Peak-to-peak ripple is 18.0V, within the 5% (20V) design constraint.

## THD Performance

- **3-Level Output:** The reduced voltage step size improves the harmonic spectrum relative to a 2-level inverter.
- **Urban (5 kHz):** Simulated current THD = 1.92%. Note: this is an optimistic bound (ideal switching model).
- **Highway (20 kHz):** Simulated current THD = 1.13%. Literature range for hardware: 1.5–4%.

## Thermal Analysis

- **Foster Network:** A 4-layer model ($R_{th,jc}$ = 0.094 °C/W) captures transient junction temperature dynamics.
- **Peak $T_j$:** Under the adaptive schedule, the junction temperature peaks at 46.4°C ($T_{ambient}$ = 40°C).
- **Safety Margin:** 128.6°C below the 175°C device limit, confirming robust thermal headroom.

# Conclusion

## Key Findings

- **Topology Validation:** The 800V 3-Level Flying Capacitor topology reduces dv/dt stress by half and improves harmonic performance relative to 2L-VSI.
- **Loss Reduction:** Adaptive switching frequency reduces urban switching losses by **75%** (84.3 → 21.1 W/phase), saving 190 W three-phase.
- **Thermal Improvement:** Thermal cycling amplitude reduced by **53%** (16.4°C → 7.7°C), extending power module fatigue lifetime per Coffin-Manson.

## Limitations & Future Work

- **Ideal Switching:** The current time-domain simulation employs ideal switching transitions.
- **Unmodeled Parasitics:** The model does not currently account for device voltage drops, discrete dead-time distortion, DC bus ripple, or stray inductance.
- **THD Caveat:** Simulated THD (1.9%, 1.1%) represents an optimistic bound; literature reports 3–8% at 5 kHz in hardware implementations.